

Davide Beltrame

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EDUCATION

Bocconi University

Milan, Italy

MSc in CS for Artificial Intelligence – on track to graduate with 110 cum laude

Sep 2024 – Oct 2026

- Main coursework: optimization and inference, information theory, computer vision, deep learning, reinforcement learning, computational neuroscience, cryptography, physics of complex systems, partial differential equations.
- Selected for the advanced course track and admitted to two PhD-level courses on statistical and applied ML.
- *Bocconi Students for Machine Learning* (BSML) Advisor & Researcher; *Bocconi Open Source Society* (BOSS) Co-Founder; Consensus Rivista Editor-in-Chief for AI and new technologies.

LUISS Guido Carli University

Rome, Italy

BSc in Management and Computer Science – **Grade:** 110/110 cum laude

Sep 2021 – Jul 2024

- Full tuition merit-based scholarship; maintained top-class ranking with 8 “laude” grades (31/30).
- **Thesis:** *ML for Eco-climatic Strategy: Predicting Air Transport Emissions*. Supervisor: Prof. Giuseppe F. Italiano.
- Starting Finance Club Guido Carli President (2023-2024); Junior Enterprise LUISS (JELU) Auditor (2022-2024).

RESEARCH EXPERIENCE

MSc Thesis Researcher

Milan, Italy

Bocconi Machine Learning & Statistical Physics Lab (Supervisor: Prof. Luca Biggio)

Sep 2025 – Present

- Studied generalization, memorization, and privacy risk in diffusion models through controlled hierarchical-data experiments and real-image benchmarks.

Student Researcher

Milan, Italy

MilaNLP Lab (Supervisors: Prof. Dirk Hovy, Dr. Paul Röttger)

Jun 2025 – Present

- Developed zero-shot length prediction for diffusion language models from latent prompt representations, mitigating the fixed-context “padding tax”; achieved significant FLOPs with minimal performance impact.

Advisor and Researcher

Milan, Italy

Bocconi Students for Machine Learning (BSML)

Apr 2025 – Present

- First-authored a study on interpretability in medical imaging, comparing saliency maps of CNNs with human radiologist attention (collected 300+ annotations from 15+ experts).
- Co-organized events on campus (HPC tutorial, meetups with industry) open to the university community.

SELECTED PUBLICATIONS

- **Co-author:** Garnier-Brun, J., Biggio, L., **Beltrame, D.**, Mézard, M., Saglietti, L. *Biased Generalization in Diffusion Models*. arXiv preprint (**recently accepted as ICML 2026 spotlight, top 2.2% of submissions**), 2026.
- **Co-first author:** Rossi, V., Cirò, G., **Beltrame, D.**, Gandolfi, L., Röttger, P., Hovy, D. *Diffusion Language Models Are Natively Length-Aware*. arXiv preprint, 2026.

WORK EXPERIENCE

AI Engineering Intern

Milan, Italy

CommerceClarity

May 2025 – Sep 2025

- Engineered an end-to-end Python pipeline for automated product image compliance, reducing the manual rejection rate from 5% to 0%, unblocking the listing funnel and eliminating human-in-the-loop review costs.
- Designed company-wide analytics dashboards (Laravel/REST/SQL) to expose operational bottlenecks and real-time ingestion metrics, adopted for weekly management reviews.

Tutor

Rome, Italy

Italian Informatics Olympics (OII) Training at LUISS

Mar 2022 – May 2023

- Architected the full technical pipeline for competitive programming contests: designed 10+ original algorithmic problems, implemented robust input generators, and wrote reference solvers in Python and C++.
- Prepared lectures on computational complexity, dynamic programming and graph theory.

SKILLS & ADDITIONAL INFORMATION

Coding & markup: Python (PyTorch, TensorFlow, Pandas, NumPy), C++, C, SQL, MATLAB, L^AT_EX, HTML.

Tools: Linux, Git, Docker, SLURM (HPC), Weights & Biases, FastAPI.

Languages: Italian (native), English (fluent C2 – IELTS 8.5), French (intermediate B2), Spanish (basic).

Volunteering: Foundation Banco Alimentare ETS (2019), APS “Il Centro” (2018-2019)

Awards: (district level) national Mathematics, Physics, Informatics and Art History Olympics.

Certifications: GRE General Test (2022): Quant 169/170, Verbal 163/170 (92nd percentile overall).

Programs: Selected for McKinsey’s *New Horizon: Lead What’s Next in AI*, Barcelona, Spain (Feb 12–14, 2026).